COHOMOLOGY OF A PROJECTIVE BUNDLE: FOR S A SCHEME, E A FINITE LOCALLY FREE O_S-MODULE OF CONSTANT RANK N + 1 AND PI : P(E) -> S, THE HIGHER DIRECT IMAGES R^Q PI_* O_{P(E)}(D) EQUAL SYM^D(E) FOR Q = 0, 0 FOR Q NOT IN {0, N}, AND SHEAFHOM(SYM^{-N-1-D}(E) TENSOR WEDGE^{N+1} E, O_S) FOR Q = N; THE IDENTIFICATIONS ARE COMPATIBLE WITH BASE CHANGE AND ISOMORPHISMS OF LOCALLY FREE SHEAVES.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2271-2298 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 2300-2412, the complete author proof (111 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-coherent (118 LIVE-clean, CONTROL.md section 2 ruling 14:3x). Top-20 triage: 9 candidates matched public corpus propositions (PUBLISHED, rejected with trace); 11 accepted here; next 9 pool ranks backfill.

Editorial note. Span computed from the \label marker through the enclosing \end{lemma}; precard_gate combined span RESULT=OK (probe L9999).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 01E8: sources/stacks-coherent/pinned-original-source.tex, source lines 2412–2412. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 015J: sources/stacks-coherent/pinned-original-source.tex, source lines 2412–2412. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01XW: sources/stacks-coherent/pinned-original-source.tex, source lines 2222–2239. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:coherent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01XT: sources/stacks-coherent/pinned-original-source.tex, source lines 1820–1823. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01XV: sources/stacks-coherent/pinned-original-source.tex, source lines 2102–2121. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0626: sources/stacks-coherent/pinned-original-source.tex, source lines 2412–2412. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $n \geq 1$. Let $\mathcal{E}$ be a finite locally free $\mathcal{O}_S$-module of constant rank $n+1$. Consider the structure morphism*

$$\pi : \mathbf{P}(\mathcal{E}) \longrightarrow S.$$

We have

$$R^q\pi_*(\mathcal{O}_{\mathbf{P}(\mathcal{E})}(d)) = \begin{cases} Sym^d(\mathcal{E}) & \text{if} \quad q = 0 \\ 0 & \text{if} \quad q \neq 0, n \\ \mathcal{H}om_{\mathcal{O}_S}(Sym^{-n-1-d}(\mathcal{E}) \otimes_{\mathcal{O}_S} \wedge^{n+1}\mathcal{E}, \mathcal{O}_S) & \text{if} \quad q = n \end{cases}$$

These identifications are compatible with base change and isomorphism between locally free sheaves.

Proof. Consider the canonical map

$$\pi^*\mathcal{E} \longrightarrow \mathcal{O}_{\mathbf{P}(\mathcal{E})}(1)$$

and twist down by 1 to get

$$\pi^*(\mathcal{E})(-1) \longrightarrow \mathcal{O}_{\mathbf{P}(\mathcal{E})}$$

This is a surjective map from a locally free rank $n+1$ sheaf onto the structure sheaf. Hence the corresponding Koszul complex is exact (More on Algebra, Lemma 0626). In other words there is an exact complex

$$0 \to \pi^*(\wedge^{n+1}\mathcal{E})(-n-1) \to \ldots \to \pi^*(\wedge^i\mathcal{E})(-i) \to \ldots \to \pi^*\mathcal{E}(-1) \to \mathcal{O}_{\mathbf{P}(\mathcal{E})} \to 0$$

We will think of the term $\pi^*(\wedge^i\mathcal{E})(-i)$ as being in degree $-i$. We are going to compute the higher direct images of this acyclic complex using the first spectral sequence of Derived Categories, Lemma 015J. Namely, we see that there is a spectral sequence with terms

$$E_1^{p,q} = R^q\pi_*\left(\pi^*(\wedge^{-p}\mathcal{E})(p)\right)$$

converging to zero! By the projection formula (Cohomology, Lemma 01E8) we have

$$E_1^{p,q} = \wedge^{-p}\mathcal{E} \otimes_{\mathcal{O}_S} R^q\pi_*\left(\mathcal{O}_{\mathbf{P}(\mathcal{E})}(p)\right).$$

Note that locally on S the sheaf $\mathcal{E}$ is trivial, i.e., isomorphic to $\mathcal{O}_S^{\oplus n+1}$, hence locally on S the morphism $\mathbf{P}(\mathcal{E}) \to S$ can be identified with $\mathbf{P}^n_S \to S$. Hence locally on S we can use the result of Lemmas 01XT, 01XV, or 01XW. It follows that $E_1^{p,q} = 0$ unless (p,q) is $(0,0)$ or $(-n-1,n)$. The nonzero terms are

$$E_1^{0,0} = \pi_*\mathcal{O}_{\mathbf{P}(\mathcal{E})} = \mathcal{O}_S$$

$$E_1^{-n-1,n} = R^n\pi_*\left(\pi^*(\wedge^{n+1}\mathcal{E})(-n-1)\right) = \wedge^{n+1}\mathcal{E} \otimes_{\mathcal{O}_S} R^n\pi_*\left(\mathcal{O}_{\mathbf{P}(\mathcal{E})}(-n-1)\right)$$

Hence there can only be one nonzero differential in the spectral sequence namely the map $d_{n+1}^{-n-1,n} : E_{n+1}^{-n-1,n} \to E_{n+1}^{0,0}$ which has to be an isomorphism (because the

spectral sequence converges to the 0 sheaf). Thus $E_1^{p,q} = E_{n+1}^{p,q}$ and we obtain a canonical isomorphism

$$\wedge^{n+1}\mathcal{E} \otimes_{\mathcal{O}_S} R^n\pi_*\left(\mathcal{O}_{\mathbf{P}(\mathcal{E})}(-n-1)\right) = R^n\pi_*\left(\pi^*(\wedge^{n+1}\mathcal{E})(-n-1)\right) \xrightarrow{d_{n+1}^{-n-1,n}} \mathcal{O}_S$$

Since $\wedge^{n+1}\mathcal{E}$ is an invertible sheaf, this implies that $R^n\pi_*\mathcal{O}_{\mathbf{P}(\mathcal{E})}(-n-1)$ is invertible as well and canonically isomorphic to the inverse of $\wedge^{n+1}\mathcal{E}$. In other words we have proved the case $d = -n-1$ of the lemma.

Working locally on S we see immediately from the computation of cohomology in Lemmas 01XT, 01XV, or 01XW the statements on vanishing of the lemma. Moreover the result on $R^0\pi_*$ is clear as well, since there are canonical maps $\mathrm{Sym}^d(\mathcal{E}) \to \pi_*\mathcal{O}_{\mathbf{P}(\mathcal{E})}(d)$ for all d. It remains to show that the description of $R^n\pi_*\mathcal{O}_{\mathbf{P}(\mathcal{E})}(d)$ is correct for $d < -n-1$. In order to do this we consider the map

$$\pi^*(\mathrm{Sym}^{-d-n-1}(\mathcal{E})) \otimes_{\mathcal{O}_{\mathbf{P}(\mathcal{E})}} \mathcal{O}_{\mathbf{P}(\mathcal{E})}(d) \longrightarrow \mathcal{O}_{\mathbf{P}(\mathcal{E})}(-n-1)$$

Applying $R^n\pi_*$ and the projection formula (see above) we get a map

$$\mathrm{Sym}^{-d-n-1}(\mathcal{E}) \otimes_{\mathcal{O}_S} R^n\pi_*(\mathcal{O}_{\mathbf{P}(\mathcal{E})}(d)) \longrightarrow R^n\pi_*\mathcal{O}_{\mathbf{P}(\mathcal{E})}(-n-1) = (\wedge^{n+1}\mathcal{E})^{\otimes -1}$$

(the last equality we have shown above). Again by the local calculations of Lemmas 01XT, 01XV, or 01XW it follows that this map induces a perfect pairing between $R^n\pi_*(\mathcal{O}_{\mathbf{P}(\mathcal{E})}(d))$ and $\mathrm{Sym}^{-d-n-1}(\mathcal{E}) \otimes \wedge^{n+1}(\mathcal{E})$ as desired. □